

ZAYAM MUSHTAQ

AI Engineer | Python & FastAPI Backend Development | Machine Learning

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PROFESSIONAL SUMMARY

Computer Science graduate (BS, University of Gujrat) with hands-on experience building AI-powered backend systems. Developed a Retrieval-Augmented Generation (RAG) pipeline using Python, FastAPI, sentence embedding, and vector search (ChromaDB) to deliver grounded, context-aware question answering. Additional experience in full-stack MERN development, REST API design, and database architecture from a final-year AI-integrated marketplace project. Seeking an AI Engineer or Machine Learning Engineer role applying practical experience in LLM integration, semantic search, and backend engineering.

TECHNICAL SKILLS

Programming Languages: Python, JavaScript, SQL

AI / Machine Learning: Retrieval-Augmented Generation (RAG), Large Language Model (LLM) Integration, Vector Databases, Semantic Search, Sentence Embedding, Hugging Face Transformers, sentence-transformers, TensorFlow.js, Natural Language Processing (NLP) Fundamentals, Prompt Design

Backend & APIs: FastAPI, Node.js, Express.js, REST API Design, Pydantic, Authentication & Authorization

Data & Databases: MongoDB, ChromaDB, Database Schema Design, Data Chunking & Retrieval

Tools & Practices: Git, Version Control, Agile/Team Collaboration, Technical Documentation

PROJECTS

RAG Knowledge Base API — Retrieval-Augmented Generation System

Built and deployed a full end-to-end RAG pipeline exposed through a FastAPI backend, combining semantic search with local LLM generation to answer questions grounded in a custom knowledge base.

- Implemented document chunking with overlap, embedding generation using sentence-transformers, and persistent vector storage with ChromaDB using cosine similarity search
- Designed a retrieval-then-generate flow that grounds LLM answers in retrieved context to reduce hallucination, running fully locally with an open-source LLM and no paid API dependency
- Exposed the system as a documented REST API (FastAPI + Pydantic) with endpoints for document ingestion, knowledge base stats, and question answering

FastAPI · Python · RAG · Sentence-Transformers · ChromaDB · Hugging Face Transformers

Petify — AI-Driven Pet Marketplace — Final Year Project, Grade: A+

Backend & Database Lead on a 3-member team that built a MERN-stack web platform featuring an AI-powered recommendation engine (TensorFlow.js) for personalized pet suggestions.

- Designed and implemented the MongoDB database schema for users, listings, and transactions
- Built backend routing and RESTful API logic using Node.js and Express.js
- Implemented authentication and data integrity safeguards to secure user and transaction data

React.js · Node.js · Express.js · MongoDB · TensorFlow.js · JWT

EDUCATION

Bachelor of Science in Computer Science — University of Gujrat, Pakistan

CGPA: 3.06 / 4.0

Relevant Coursework: Database Management, Web Technologies, Business Communication, Documentation & Reporting, Team Collaboration